Allouche Mohamed Abderrahmane

Full-Stack Software Engineer

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Professional Summary

Software Engineer specializing in full-stack development with TypeScript, React, Next.js, Node.js, Python, and PostgreSQL. Experienced in designing REST APIs, reusable frontend components, relational databases, and asynchronous services. Builds secure applications with OAuth 2.0, JWT authentication, and role-based access control, using Docker and Git throughout development and deployment. Combines hands-on engineering with end-to-end feature ownership, debugging, and code reviews. Computer Science graduate ranked third in a cohort of 120+, currently studying Cybersecurity at WSB University in Kraków.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Python, SQL, Java, PHP, HTML5/CSS3, C# , GO
Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Bootstrap, Vite, Responsive UI , React Native
Backend: FastAPI, Node.js, Express.js, Django, ASP.NET CORE, Spring Boot Pydantic, SQLAlchemy, Celery
Databases: PostgreSQL, Supabase, MongoDB, MySQL
APIs & Authentication: REST APIs, WebSockets, JWT, OAuth 2.0, Role-Based Access Control (RBAC), Argon2
DevOps & Tools: Docker, Git, GitHub, Vercel, Railway, Render, Linux
Security: OWASP Top 10, Wazuh, Burp Suite, Metasploit, Nmap, Wireshark, Kali Linux

Work Experience

FlyRank Corp. (FlyRank AI) Remote, USA
Back-End AI Engineering Intern July 2026 – sep 2026

- Engineered backend RESTful API endpoints in Python and FastAPI during a 6-week intensive engineering program to support AI-driven features and model integration workflows.
- Architected asynchronous data-processing pipelines leveraging background task queues, Pydantic request validation, and structured JSON serialization for model outputs.
- Partnered with cross-functional engineering teams to formulate clean backend architecture, relational database schemas, API contracts, and technical documentation.

SocShield DZ Hybrid, Algeria
Full-Stack Developer Aug 2024 – Jan 2026

- Constructed modular security-monitoring dashboards across 15+ data views using React, TypeScript, and Redux Toolkit, managing complex asynchronous application state.
- Designed and deployed 20+ RESTful API endpoints using FastAPI and PostgreSQL, implementing Pydantic validation schemas and SQLAlchemy for relational database querying.
- Orchestrated token-based JWT authentication and Role-Based Access Control (RBAC) across 4 user privilege tiers, enforcing strict authorization policies across client portals.
- Enforced OWASP Top 10 security standards across application layers, mitigating injection and XSS risks through parameterized database queries and structured code reviews.

Projects

LeakShield – Cybersecurity SaaS Platform | *FastAPI, SQLModel, PostgreSQL, Celery, React*

- Spearheaded backend infrastructure for an automated OSINT data-breach monitoring platform, orchestrating Celery workers and PostgreSQL to index exposed digital assets.
- Created responsive React dashboards with real-time search, multi-criteria filtering, and asset-exposure synchronization backed by Supabase.

Devfolio – AI Portfolio Builder | *FastAPI, React, PostgreSQL, Pydantic, SQLAlchemy*

- Authored a full-stack portfolio platform using FastAPI, React, and PostgreSQL, enabling users to upload resume files and generate customized, responsive personal websites.
- Integrated third-party GitHub OAuth 2.0 and local authentication secured with memory-hard Argon2 password hashing and JWT access tokens.

Pulse-Chat – Real-Time Messaging Application | *React, TypeScript, FastAPI, WebSockets*

- Engineered a full-duplex real-time messaging application using React, TypeScript, FastAPI, and WebSockets, sustaining 100+ concurrent connections in testing.
- Designed multi-room messaging architecture featuring authenticated WebSocket handshakes, connection lifecycle monitoring, and automated client reconnection logic.

Education

WSB University — Kraków, Poland
B.Sc. in Cybersecurity — Expected: Oct 2026 – July 2030

Université Dr. Moulay Tahar — Saïda, Algeria
B.Sc. in Computer Science – Ranked 3rd in Cohort of 120+ — Sept 2023 – June 2026

Relevant Coursework: Data Structures & Algorithms, Web Development, Web Security, Operating Systems, Linux Systems, AI & Machine Learning, Language Theory & Compilation.

Achievements & Certifications

Achievements: 2nd Place Winner – DevRush 1.0 National Programming Competition (2025); National Finalist – FORSA Algérie Télécom Hackathon & Datathon (2025)

Certifications: Google Cybersecurity Professional Certificate (2024); NVIDIA Fundamentals of Deep Learning (2025); Huawei HCIA-Datacom (2025); Simplilearn CCNA Network Associate (2025)

Languages: Arabic (Native), English (Professional – IELTS Academic 6.5, EFSET C1), French (Advanced – TCF C1)